



US012406133B2

(12) **United States Patent**
Li et al.

(10) **Patent No.:** **US 12,406,133 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) **METHOD AND APPARATUS FOR
DISPLAYING TEXT CONTENT COPIED
FROM A FIRST APPLICATION IN A
SECOND APPLICATION**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/402,962**

(22) Filed: **Jan. 3, 2024**

(65) **Prior Publication Data**

US 2024/0143904 A1 May 2, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/099,680, filed on
Nov. 16, 2020, now Pat. No. 11,868,710, which is a
(Continued)

(51) **Int. Cl.**
G06F 40/174 (2020.01)
G06F 16/3329 (2025.01)

(Continued)

(52) **U.S. Cl.**
CPC **G06F 40/174** (2020.01); **G06F 16/3329**
(2019.01); **G06F 16/3334** (2019.01);
(Continued)

(58) **Field of Classification Search**
CPC G06F 40/174; G06F 40/166; G06F 40/30;
G06F 16/3329; G06F 16/3334; G06F
16/35

See application file for complete search history.

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(57) **ABSTRACT**

A method is applied to an electronic device, and includes:
obtaining content copied by a user on a first page, and
determining a category to which the content belongs; obtain-
ing a category of a second page; matching the category of
the content against the category of the second page; and if
a matching result is not empty, displaying at least one
operation option on the second page, where the at least one
operation option has a function of performing an operation
for the content on the second page, so that the electronic
device directly performs the function after the user selects
the operation option.

24 Claims, 6 Drawing Sheets

Electronic map
Search for a location or route
Map
Locate: Nanjing South railway station
Navigate to: Nanjing South railway station
Ignore

Related U.S. Application Data

continuation of application No. 16/097,263, filed on Oct. 28, 2018, now Pat. No. 10,853,564, which is a continuation of application No. PCT/CN2016/080438, filed on Apr. 28, 2016.

(51) Int. Cl.

G06F 16/3332 (2025.01)

G06F 16/35 (2019.01)

G06F 40/166 (2020.01)

G06F 40/30 (2020.01)

(52) U.S. Cl.

CPC **G06F 16/35** (2019.01); **G06F 40/166** (2020.01); **G06F 40/30** (2020.01)

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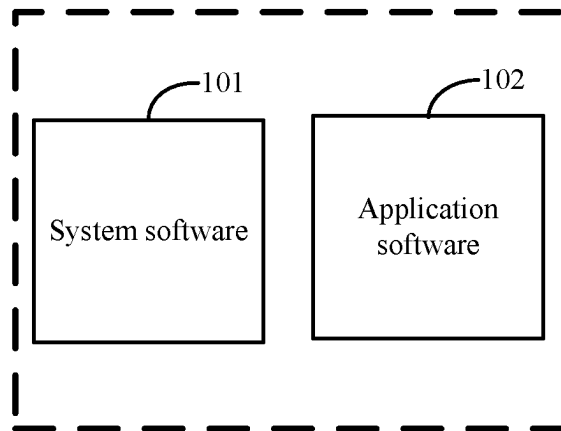


FIG. 1

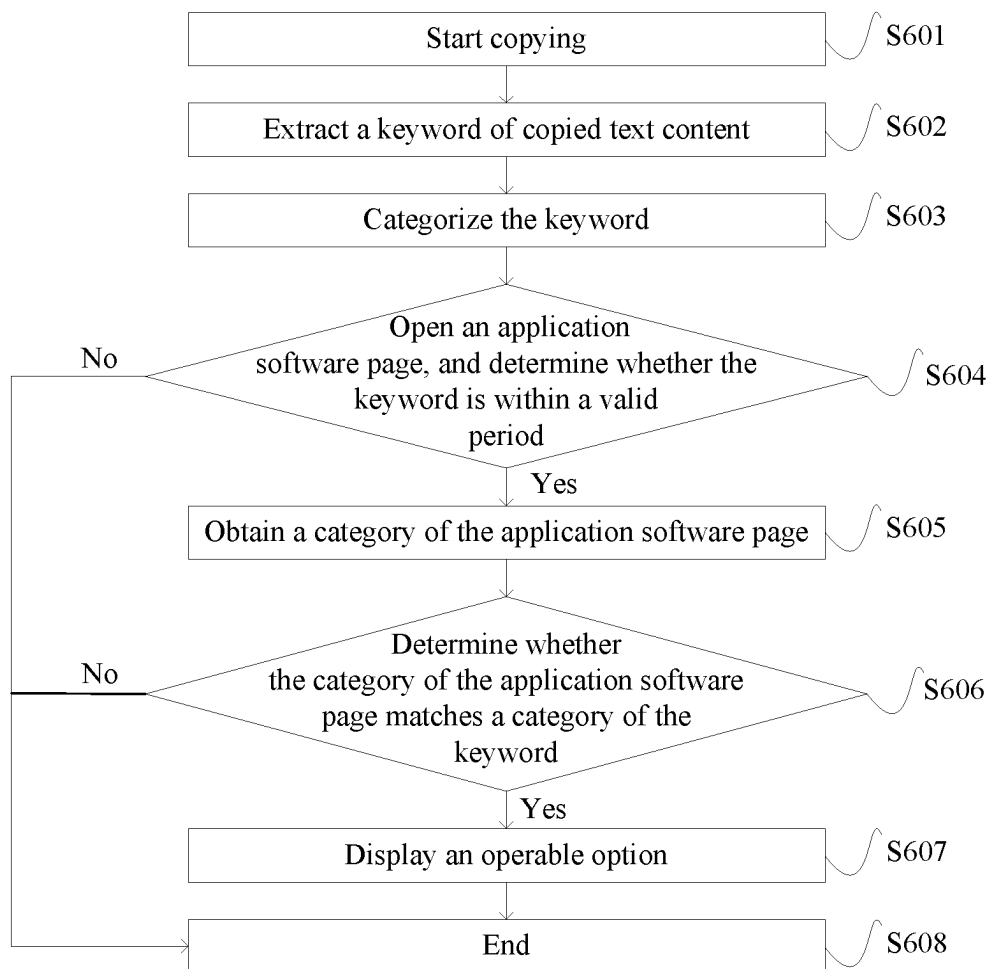


FIG. 2

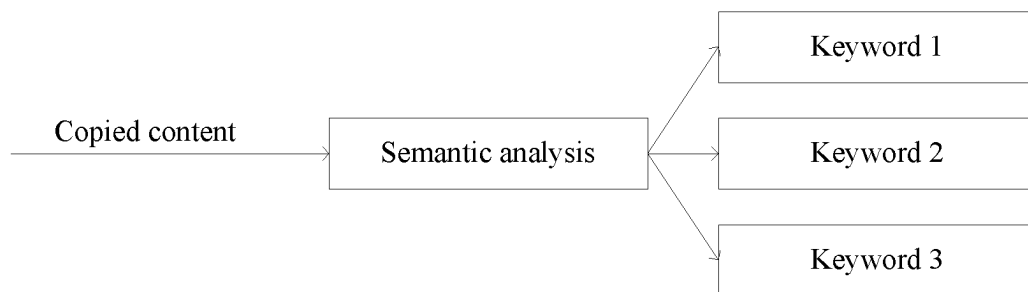


FIG. 2a

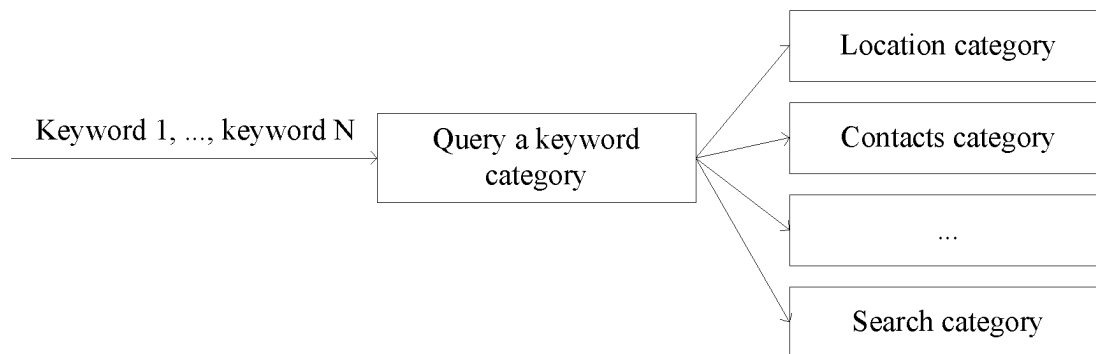


FIG. 2b

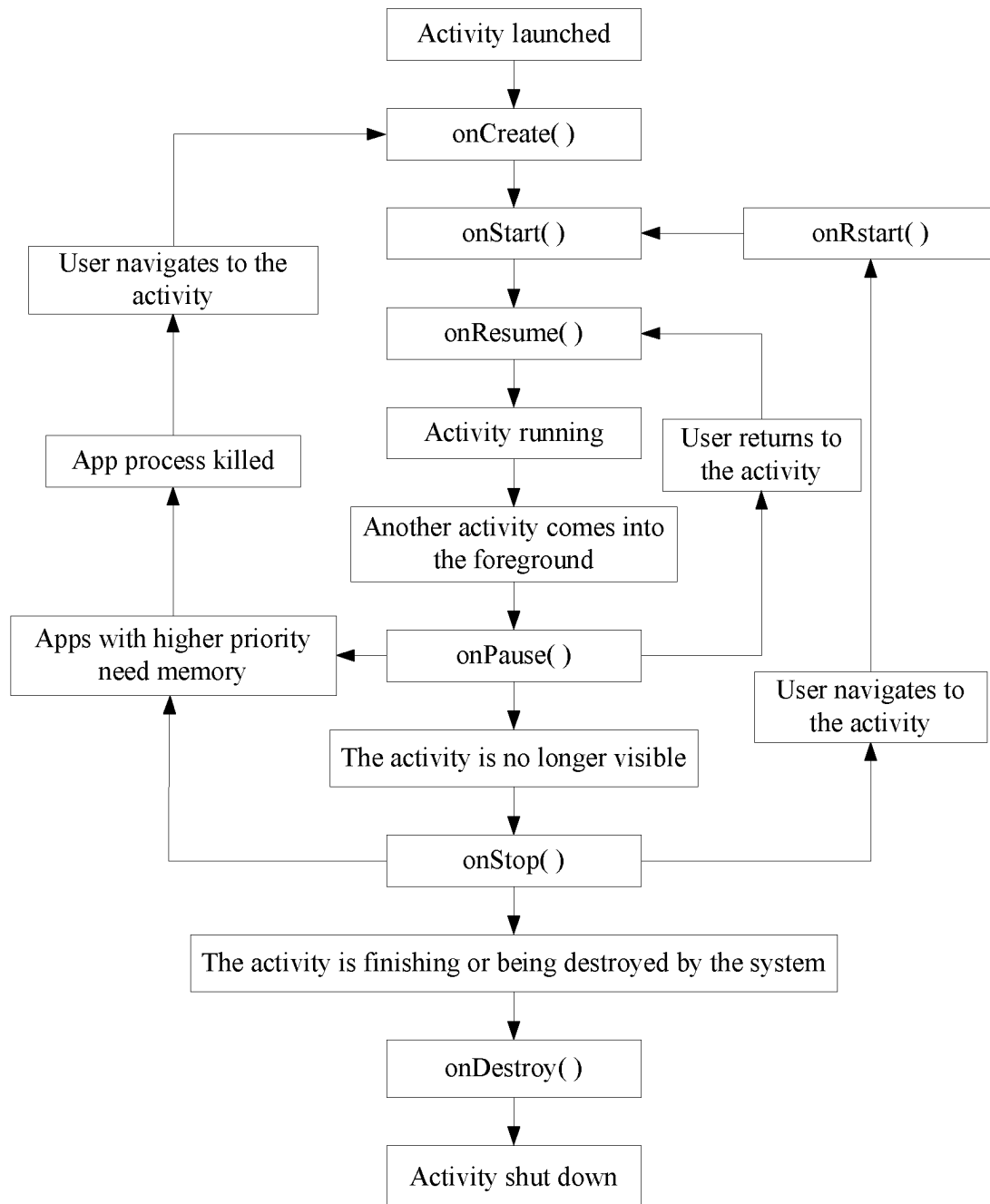


FIG. 2c

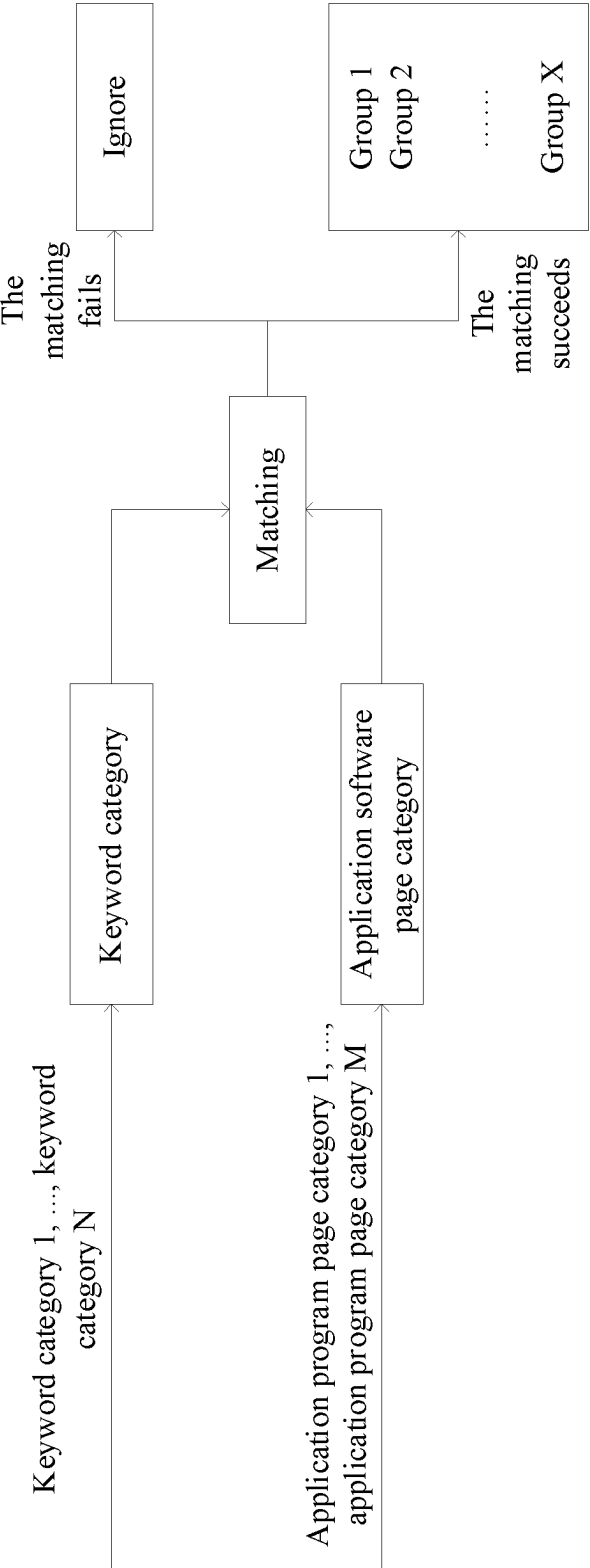


FIG. 2d

Electronic map

Search for a location or route

Map

Locate: Nanjing South railway station

Navigate to: Nanjing South railway station

Ignore

FIG. 2e

Address book

Search for a contact

Contacts information

Add a contact:
Mike/12312345678

Ignore

FIG. 3

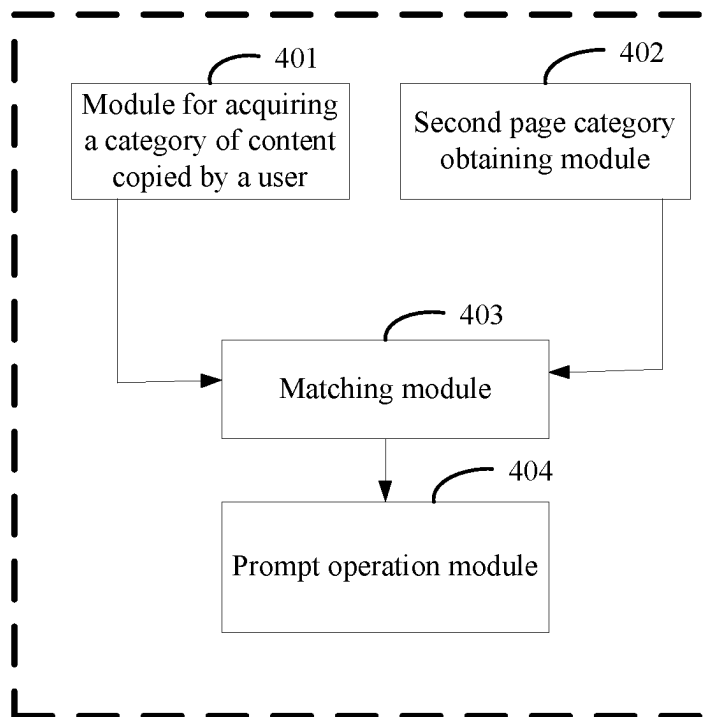


FIG. 4

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METHOD AND APPARATUS FOR DISPLAYING TEXT CONTENT COPIED FROM A FIRST APPLICATION IN A SECOND APPLICATION

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/099,680, filed on Nov. 16, 2020, which is a continuation of U.S. patent application Ser. No. 16/097,263, filed on Oct. 28, 2018, now U.S. patent Ser. No. 10/853,564. The U.S. patent application Ser. No. 16/097,263 is a continuation of International Application No. PCT/CN2016/080438, filed on Apr. 28, 2016. All of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

The present invention relates to the field of mobile communications, and in particular, to a human-computer interaction method and an apparatus thereof.

BACKGROUND

At present, existing smartphones all have basic copy and paste functions. A user performs a copy operation on a panel having content that needs to be copied, to copy the content into a clipboard; and then opens another application or switches to another page, chooses a target area, and long presses the target area to paste the copied content before the user can finally perform a desired further operation.

After the user opens these applications or switches to these pages, an expected to-be-performed action is usually predictable. However, the user needs to always perform a plurality of repeated operations to complete the expected action. For example, the user copies "Nanjing South railway station" to the clipboard, and opens a map application to enter a map page, and the user needs to choose an address input box first, and long press the address input box for pasting before the user can further tap a search option to view desired information. This undoubtedly burdens the user with some unnecessary trivial operations.

In the prior art, using locating on an electronic map as an example, an entire copy-paste procedure includes the following steps: selecting desired to-be-copied text; then tapping the text and copying the text; opening an application software page; choosing an input box, and long pressing the input box for pasting; and pasting the text into the input box of the electronic map to perform searching and locating.

Disadvantages of the foregoing paste manner in the prior art are as follows: First, the user needs to additionally tap the input box (a target area), long press the input box, wait until a paste option is displayed, and tap the paste option for pasting before the user can finally search for target information. Second, the user probably needs to perform tapping a plurality of times to choose the input box; unintentional sliding may occur during long pressing after the input box is chosen, and the user needs to long press the input box again. Moreover, the user is likely to forget the copied content and needs to perform a plurality of operations before the user can confirm the copied content.

SUMMARY

To overcome disadvantages in the prior art, the present invention provides a human-computer interaction method and an apparatus thereof. The method can reduce complexity of user operations.

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A first aspect of the present invention provides a human-computer interaction method, applied to an electronic device. The method includes: obtaining content copied by a user on a first page, and determining a category to which the content belongs; obtaining a category of a second page; matching the category of the content against the category of the second page; and if a matching result is not empty, displaying at least one operation option on the second page, where the at least one operation option has a function of performing an operation for the content on the second page, so that the electronic device directly performs the function after the user selects the operation option.

In a first possible implementation of the first aspect, the determining a category to which the content belongs includes: performing semantic analysis on the copied text content to extract a keyword of the copied content, and matching the keyword against a keyword category library to obtain the category to which the content belongs.

With reference to the first possible implementation of the first aspect, in a second possible implementation, the copied text content includes one or more keywords, and there are one or more categories of the keyword.

In a third possible implementation of the first aspect, there is one or more categories of the second page.

With reference to the third possible implementation of the first aspect, in a fourth possible implementation, one or more categories to which the content belongs are separately matched against one or more categories of the second page.

In a fifth possible implementation of the first aspect, before the step of obtaining a category of a second page, the method further includes: determining that a time interval between the copy operation of the user and opening of the second page is within a preset time limit range.

With reference to the first, second, third, fourth, or fifth possible implementation of the first aspect, in a sixth possible implementation, the copied content is text information, the copied text content is text information rather than image, sound or video data, and common text document name extensions include .txt, .doc, .docx, .wps, and the like.

In a seventh possible implementation of the first aspect, the category of the second page is predefined, and categorization of the second page is consistent with categorization of the keyword, so that it is more convenient to match the category of the second page against the category of the keyword.

A second aspect of the present invention provides a human-computer interaction apparatus. The apparatus includes: a module for obtaining a category of content copied by a user, a second page category obtaining module, a matching module, and a prompt operation module, where the module for obtaining a category of content copied by a user is configured to: obtain content copied by a user on a first page, and determine a category to which the content belongs; the second page category obtaining module is configured to obtain a category of a second page; the matching module is configured to match the category of the content against the category of the second page; and the prompt operation module is configured to: if a matching result is not empty, display at least one operation option on the second page, where the at least one operation option has a function of performing an operation for the content on the second page, and the function is directly performed after the user selects the operation option.

In a first possible implementation of the second aspect, the module for obtaining a category of content copied by a user is configured to: perform semantic analysis on the text content copied by the user to obtain a keyword of the copied

text content, and match the keyword against a keyword category library to extract a category of the keyword.

In a second possible implementation of the second aspect, the matching module is configured to separately match one or more categories of the keyword against one or more categories of the second page, to obtain one or more matched combinations of the keyword and the second page.

The apparatus includes a mobile phone, a tablet computer, a palmtop computer, and a personal digital assistant (Personal Digital Assistant, PDA).

In embodiments of the present invention, a category of copied text content that is extracted is matched against a category of a keyword, and a matched group is used as a prompt operation. The prompt operation is displayed in a form of a dialog box, to prompt a next action that may be performed by a user, so that complexity of user operations is reduced, and accuracy of pasting text information is improved.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a block diagram of smart paste according to an embodiment of the present invention;

FIG. 2 is a schematic flowchart of a smart paste method according to an embodiment of the present invention;

FIG. 2a is a schematic flowchart of extracting a keyword according to an embodiment of the present invention;

FIG. 2b is a schematic diagram of matching a keyword against a keyword category library according to an embodiment of the present invention;

FIG. 2c is a schematic diagram of a lifecycle of each activity maintained by an Android system according to an embodiment of the present invention;

FIG. 2d is a schematic diagram of matching a keyword category against an application software page category according to an embodiment of the present invention;

FIG. 2e is a schematic diagram of prompt operations displayed on an electronic map according to an embodiment of the present invention;

FIG. 3 is a schematic diagram of a prompt operation displayed in an address book according to an embodiment of the present invention; and

FIG. 4 is a schematic structural diagram of a human-computer interaction apparatus according to an embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

The following further describes the technical solutions of the present invention in detail with reference to the accompanying drawings and embodiments.

In the embodiments of the present invention, when a user copies text, a keyword in copied text content is categorized. When the user enters an application software page, a category of the application software page is obtained, and is matched against a category of the keyword. Finally, a matched category is used as a prompted operation option, and the prompted operation option is displayed in a form of a dialog box. The user performs a tap operation on the prompted operation option. Therefore, problems of complex operations, low accuracy, and trivial operations in a copy-paste method in the prior art are resolved.

The method in the embodiments of the present invention is applied to system software (the system software is a system supporting development and running of application software, and is a set of various programs that do not need to be intervened with by a user) of a terminal. The terminal

includes pieces of application software (the application software is a set of application programs that are compiled by using various programming languages, and is a computer software program designed to resolve various application problems, such as a navigation app or Word).

The terminal may include a processor, a memory, an input module, a display module, a network communications module, and a system bus. The system bus is configured to perform information transfer between components of the terminal. The input module may include one or more of components such as a keyboard, a touchscreen, and a mouse, and is configured to receive information entered by a user and send the received information to the processor for processing. The display module is configured to display content that the terminal needs to display. The input module and the display module may be integrated in one touchscreen. The network communications module is configured to implement a function of communication between the terminal and an external device. The memory may include one or more computer readable storage media, and may store corresponding programs such as an operating system and application software. For example, if a navigation app is installed on the terminal, application software used to install the navigation app is stored in the memory of the mobile phone terminal. The processor runs software programs and modules stored in the memory to perform various functional applications of the terminal and perform data processing.

The following uses FIG. 1 as an example for description. FIG. 1 is a block diagram of smart paste according to an embodiment of the present invention, and the block diagram includes system software 101 and application software 102. In this embodiment of the present invention, the system software 101 is developed based on an operating system of a terminal, and the application software 102 on the terminal is developed based on the system software 101. After a user copies text, a category of a keyword COPYKEY in copied text content is matched against a category of a page of the application software 102. Because one piece of application software 102 includes a plurality of application software pages, and a page category ACTIVITYTYPE is preset in an attribute of an application software page. In other words, a category attribute TYPE is added in an application software page ACTIVITY. Therefore, when the system software 101 runs, the category ACTIVITYTYPE of the page of the application software 102 may be directly read and matched against a category COPYTYPE of the keyword. Finally, a matched category is used as an operation option and the operable operation is displayed in a dialog box, to complete an entire copy-paste process. The paste described herein is not an operation of pasting text that is temporarily stored in a clipboard in the prior art, but instead, is a process in which the system software 101 displays, in a dialog box, the matched category for the user to select.

It should be noted that in implementation of the present invention, the terminal may be a terminal device that has a function of installing application software, such as a mobile phone, a tablet computer, a palmtop computer, or a personal digital assistant (Personal Digital Assistant, PDA). This is not limited.

The present invention provides a human-computer interaction method, applied to an electronic device, and including:

- obtaining content copied by a user on a first page, and determining a category to which the content belongs;
- obtaining a category of a second page;
- matching the category of the content against the category of the second page; and

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if a matching result is not empty, displaying at least one operation option on the second page, where the at least one operation option has a function of performing an operation for the content on the second page, so that the electronic device directly performs the function after the user selects the operation option.

This embodiment of the present invention is based on an Android operating system. The first page and the second page are both application software pages. The first page and the second page may be a same application software page or may be different application software pages. The following uses an electronic map app as an example for description. FIG. 2 is a schematic flowchart of a smart paste method according to an embodiment of the present invention. As shown in FIG. 2, the method includes steps S601 to S608.

Step S601: Start copying.

A user chooses to copy text on an application software page of a terminal, where copied text content is temporarily stored in a clipboard.

It should be noted that one piece of application software in the terminal may include one or more application software pages. For example, in an electronic map, there is an application software page for locating, an application software page for navigation, and further an application software page for viewing real-time traffic, which are collectively referred to as application software pages in this embodiment of the present invention. The application software includes an electronic map app, an electronic address book app, a browser app, and the like, which are not listed one by one herein.

In this embodiment of the present invention, the copied text is a document type in a computer, and is mainly used to record and store text information rather than image, sound or video data. Common text document name extensions include .txt, .doc, .docx, .wps, and the like, which are not limited and are collectively referred to as text.

Step S602: Extract a keyword of copied text content.

As shown in FIG. 2a, FIG. 2a is a schematic flowchart of extracting a keyword according to an embodiment of the present invention. After semantic analysis is performed on the text content copied by the user, N (N is an integer greater than or equal to 1) keywords are extracted from the copied content.

For example, a keyword “Nanjing South railway station” is extracted from copied text content “Passengers can take Nanjing Metro Line 1 to Nanjing South railway station”. A plurality of keywords may be extracted from the copied text content, and keyword indexes are established. The extracted keyword is shown in Table 1. Table 1 is a schematic diagram obtained after a keyword is extracted according to an embodiment of the present invention.

TABLE 1

Index	Keyword
1	Nanjing South railway station

It should be noted that when the user copies the text content, the user may choose to directly copy “Nanjing South railway station”, and semantic analysis is then performed on the copied text content “Nanjing South railway station” to obtain the keyword “Nanjing South railway station”. To be specific, semantic analysis is performed for purposes of a more convenient and accurate process of obtaining the keyword from the copied text content. For example, when the user performs a copy operation, the

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keyword “Nanjing South railway station” can be obtained through semantic analysis even if the copied text content is “To Nanjing South railway station”.

Step S603: Analyze and categorize the keyword.

As shown in FIG. 2b, FIG. 2b is a schematic diagram of matching a keyword against a keyword category library according to an embodiment of the present invention.

The extracted keyword is matched against the keyword category library to determine a category of the keyword.

It should be noted that the keyword category library is stored in a memory of the terminal. A path of the keyword category library is written into a program of system software in this embodiment of the present invention. When the program of the system software runs, a processor directly invokes the keyword category library to match the extracted keyword against the keyword category library. The keyword category library is preset and supports expansion. The keyword category library includes a location category, a contacts category, a search category, a web address category, a text path category, a video resource path category, an audio resource path category, and the like. In each category of the keyword, a plurality of attributes that belong to characteristics of the category are prestored, for example, “Nanjing South railway station”, “Beijing West railway station”, and the like are stored in the location category.

In this embodiment of the present invention, each keyword may belong to a plurality of categories. For example, a keyword “Nanjing South railway station” belongs to a location category to perform navigation or locating on “Nanjing South railway station”, and further belongs to a search category to perform a search on a browser page. After a category of the keyword is determined, as shown in Table 2, Table 2 is a schematic diagram obtained after a keyword is matched against the keyword category library according to an embodiment of the present invention. Table 2 shows one keyword “Nanjing South railway station”, and two keyword categories, which are a keyword category 1 “location category” and a keyword category 2 “search category” respectively.

TABLE 2

Index	Keyword	Keyword category 1	Keyword category 2
1	Nanjing South railway station	Location category	Search category

Step S604: Open an application software page, and determine whether the keyword is within a valid period.

The user copies text, and then opens an application software page after a particular time. In other words, there is a period of time between copying of the text and an operation of opening application software. When the user opens the application software, for a keyword temporarily stored in a clipboard, the program of the system software needs to determine whether a time interval between the copying of the text and the operation of opening the application software is within a predetermined time.

If a determining result is that the time interval between the copying of the text and the operation of opening the application software is within the predetermined time or is equal to the predetermined time, step S605 is performed. If a determining result is that the time interval between the copying of the text and the operation of opening the application software exceeds the predetermined time, step S605 is not performed, and therefore step S608 is performed.

In this embodiment of the present invention, the predetermined time is a time set by default by the program of the system software. The user may customize the predetermined time based on a requirement. For example, the time interval between the copy operation and the operation of opening the application software is set to five minutes by using a settings menu bar of a mobile phone terminal. In this case, after the copy operation is completed, a valid period of the copied text is five minutes. If the time interval between the copy operation and the operation of opening the application software exceeds five minutes, the copied text becomes invalid and subsequent steps can no longer be performed, but step S608 is directly performed instead.

It should be noted that compared with the duration of step S603 and step S604, the duration of step S601 to step S603 is negligible. The time interval between the copy operation and the operation of opening the application software described in this embodiment of the present invention is a time between step S603 and step S604. For example, between step S603 and step S604, the user has not opened application software for a long time after copying the text, but instead chooses to power off the terminal. In this case, the copied text becomes invalid.

Step S605: Obtain a category of the application software page.

The application software opens an interface to a system software vendor or a terminal device manufacturer or supports a system architecture such as an Intent architecture of Android to implement that the terminal device obtains and uses a category of an application software page.

Categories of application software pages include a location category, a contacts category, a search category, a web address category, a text path category, a video resource path category, an audio resource path category, and the like.

Application software pages and categories of the application software pages are shown in Table 3. Table 3 is a schematic diagram of application software pages and categories of the application software pages according to an embodiment of the present invention.

TABLE 3

Application program page	Application program page category
Electronic map page	Location category
Browser page	Search category

It should be noted that the categories of the application software pages are consistent with categories in the keyword category library to facilitate matching in step S606. Because a category attribute TYPE is added to an application software page ACTIVITY, the program of the system software reads a category ACTIVITYTYPE of the application software page. However, when a category attribute TYPE is not added to an application software page ACTIVITY, the program of the system software obtains a category of application software and uses the category of the application software as the category of the application software page. The obtaining a category of application software herein is as follows: Application software apps whose user quantities rank on top in markets are categorized in advance to obtain a category table, and the category table is stored on a server or a cloud, and is stored into a memory of a terminal during installation of the application software app on the terminal.

Alternatively, a terminal may obtain a category of application software from a server or a cloud when querying the category of the application software. Alternatively, a user may manually add a category to application software. When the user manually adds a category, a system may prompt the user to select and add a category that can be added. Alternatively, a category of application software may be obtained from another terminal, for example, category information of application software is obtained in a manner such as Bluetooth or WLAN. When the program of the system software performs step S605, the category table is directly read from the memory of the terminal, or the category of the application software is obtained from a network side, or information is obtained from another terminal, so as to obtain the category of the application software. For example, even if TYPE is not defined on an application software page for the Baidu Maps app, the AutoNavi map app, the Sogou Maps app, and the like, these apps can be classified into a location category to obtain a category table for storage.

Step S606: Determine whether the category of the application software page matches a category of the keyword.

A category of an application software page, after being obtained, needs to be matched against a category of a keyword extracted from copied text. The matching means searching one or more categories one by one for the keyword extracted from the copied text to determine whether the category of the keyword is consistent with the category of the application software page.

It should be noted that an application software page and a keyword each may belong to more than one category. Therefore, a two-layer cyclic manner needs to be used to perform comparison on each combination.

The following uses FIG. 2c as an example to describe a specific embodiment of obtaining a category of an application software page. Specifically, FIG. 2c shows a lifecycle of each activity maintained by an Android system. When an Activity is entered for the first time, an onCreate() method of the Activity is performed. Operations of obtaining a category of the Activity and matching the category of the Activity against content in a clipboard may be added to an onCreate() method of an Activity.java class.

A category of each Activity is configured in an Android-Manifest.xml file as follows:

```

<activity
  android:name="com.example.smartpaste.MainActivity"
  android:label="@string/app_name" >
  <meta-data
    android:name="sort"
    android:value="search|navigation|location" />
  </intent-filter>
</activity>

```

A category is configured in a form of a key-value (key-value) pair. A plurality of values may be configured in "value", and are separated by "|".

```

Modify Activity.onCreate( ):
protected void onCreate(Bundle savedInstanceState) {
    ...
    String sort[ ] = readCategory( );//Read a category of an Activity
    matchClipboard(sort);//Match a category against a keyword in a clipboard
    ...
}
private String[ ] readCategory( ) {
    try {
        ActivityInfo activityInfo = this.getPackageManager( ).getActivityInfo(
            getComponentName( ),
            PackageManager.GET_META_DATA);
        String value = activityInfo.metaData.getString("sort", null);
        if (!TextUtils.isEmpty(value)) {
            String sort[ ] = value.split("\\|");
            return sort;
        }
    } catch (NameNotFoundException e) {
        e.printStackTrace( );
    }
    return null;
}
private void matchClipboard(String[ ] sort) {
    //TODO
}

```

Callback after an option is tapped:

//Callback, a callback after "Search for xxx" is tapped:

```
public void onSmartPasteForSearch(String value) { }
```

//Callback, a callback after "Locate xxx" is tapped

```
public void onSmartPasteForLocation(String loc) { }
```

Callback, a callback after "Navigate to xxx" is tapped

```
public void onSmartPasteForNavigation(String loc) { }
```

The following uses FIG. 2d as an example for description. FIG. 2d is a schematic diagram of matching a category of a keyword against a category of an application software page according to an embodiment of the present invention.

As shown in FIG. 2d, a keyword extracted from copied text may belong to a plurality of categories, which are N (N is an integer greater than 0) keyword categories. An application software page may also belong to a plurality of categories, which are M (M is an integer greater than 0) application software page categories. Each category of the keyword needs to be matched against each category of the application software page. If the matching fails, the combination for matching is ignored, and step S608 is performed. If the matching succeeds, all matched combinations are stored, the matched combinations are stored in a buffer allocated by a processor of a terminal to system software in this embodiment of the present invention, and step S607 is performed. There may be a plurality of matched combinations, which are X matched groups (X is a positive integer greater than 0).

Matching results are shown in Table 4. Table 4 is a schematic diagram obtained after categories of a keyword are matched against categories of application software pages according to an embodiment of the present invention.

TABLE 4

Application program page	Application program page category	Keyword category	Matching result
Electronic map page	Location category	Location category	Matching succeeds

TABLE 4-continued

Application program page	Application program page category	Keyword category	Matching result
Electronic map page	Location category	Search category	Matching fails
Browser page	Search category	Search category	Matching succeeds

Step S607: Display an operable option.

After the category of the application software page matches the category of the keyword, there may be a plurality of X matched groups. In this case, all matched groups are displayed. In other words, a "paste" operation is completed.

Table 5 is a schematic diagram obtained after a keyword matches application software pages according to an embodiment of the present invention.

TABLE 5

Application program page	Keyword	Keyword category	Prompt operation
Electronic map page	Nanjing South railway station	Location category	Position: Nanjing South railway station
Electronic map page	Nanjing South railway station	Location category	Navigate to: Nanjing South railway station
Browser page	Nanjing South railway station	Search category	Search: Nanjing South railway station

As shown in Table 5, categories of a keyword "Nanjing South railway station" include a location category and a search category. An electronic map page belongs to the location category. After the matching succeeds, the matched group is displayed in a form of a dialog box. To be specific, during a prompt operation, "Locate: Nanjing South railway station" and "Navigate to: Nanjing South railway station" are displayed. A browser page belongs to a search category, and after the matching succeeds, the matched group is displayed in a dialog box. To be specific, during a prompt operation, "Search for: Nanjing South railway station" is displayed.

This is described below by using an electronic map as an example. FIG. 2e is a schematic diagram of prompt opera-

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tions displayed on an electronic map of a terminal according to an embodiment of the present invention.

As shown in FIG. 2e, a matched group is displayed on an electronic map. To be specific, a location category of a keyword “Nanjing South railway station” matches a location category of an electronic map page. The electronic map page belongs to a location category, but a category of an electronic map is more accurately classified into “Navigate” and “Locate”. Therefore, after a user opens the electronic map page, “Locate: Nanjing South railway station” and “Navigate to: Nanjing South railway station” are displayed. Apparently, it is unlikely herein to display all operations that the user wants to perform. In other words, the user may want to perform an operation other than “Navigate” or “Locate”. Therefore, an “Ignore” option is provided in the displayed dialog box for the user to select. The user can directly perform the operation of navigating to or locating Nanjing South railway station by tapping “Navigate” or “Locate”.

The following describes the smart paste method in this embodiment of the present invention by using address book application software as an example.

A user copies a piece of text information “Xiao ming’s phone number is 12312345678” on a page of application software. A program of system software performs semantic analysis on the copied text information to obtain a keyword “Xiao ming/12312345678”. In this case, the program of the system software invokes a keyword category library, to match the keyword against the keyword category library, and it is obtained that the category of the keyword is “contacts category”. When the user opens a page of the address book application software, a valid period of the keyword temporarily stored in a clipboard is determined. If a time interval between the copy operation and the page opening operation exceeds the valid period, running of the program of the system software is directly terminated. If the time interval is within the valid period, it is obtained that a category of the page of the address book application software is “contacts category”, and the obtained “contacts category” of the keyword is matched against the category of the page of the address book application software, to obtain a matching result “Contacts category matching succeeds”.

As shown in FIG. 3, FIG. 3 is a schematic diagram of a prompt operation displayed on an address book according to an embodiment of the present invention. A dialog box is displayed on a page of the address book application software. Operable options “Add a contact: Xiao ming/12312345678” and “Ignore” are displayed for the user to select. The user can directly perform an operation of adding a contact by tapping “Add a contact: Xiao ming/12312345678”.

In this embodiment of the present invention, a category of copied text content is extracted and is matched against a category of a keyword, and a matched group is used as a prompt operation. The prompt operation is displayed in a form of a dialog box, to intelligently push a next action that may be performed by a user, so that complexity of user operations is reduced, and accuracy of pasting text information is improved.

The following uses a smart terminal as an example to describe embodiments of the present invention. FIG. 4 is a schematic structural diagram of a human-computer interaction apparatus according to an embodiment of the present invention.

As shown in FIG. 4, the apparatus includes a module 401 for obtaining a category of content copied by a user, a second page category obtaining module 402, a matching module 403, and a prompt operation module 404.

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The module 401 for obtaining a category of content copied by a user is configured to: obtain content copied by a user on a first page, and determine a category to which the content belongs.

The second page category obtaining module 402 is configured to obtain a category of a second page.

The matching module 403 is configured to match the category of the content against the category of the second page.

The prompt operation module 404 is configured to: if a matching result is not empty, display at least one operation option on the second page, where the at least one operation option has a function of performing an operation for the content on the second page, and the function is directly performed after the user selects the operation option.

In the apparatus, the module 401 for obtaining a category of content copied by a user is further configured to: perform semantic analysis on the text content copied by the user to obtain a keyword of the copied text content, and match the keyword against a keyword category library to extract a category of the keyword.

In the apparatus, the matching module 403 is further configured to: separately match one or more categories of the keyword against one or more categories of an application software page, to obtain one or more matched combinations of the keyword and the application software page.

In the apparatus, a matched combination of a category of a keyword and a category of an application software page is used as an operable option and is displayed in a form of a dialog box, so that a “paste” operation in this embodiment of the present invention is completed.

In this embodiment of the present invention, a category of copied text content is extracted and is matched against a category of a keyword, and a matched group is used as a prompt operation. The prompt operation is displayed in a form of a dialog box, to intelligently push a next action that may be performed by a user, so that complexity of user operations is reduced, and accuracy of pasting text information is improved.

A person skilled in the art may be further aware that, in combination with the examples described in the embodiments disclosed in this specification, units and algorithm steps may be implemented by electronic hardware, computer software, or a combination thereof. To clearly describe the interchangeability between the hardware and the software, the foregoing has generally described compositions and steps of each example according to functions. Whether the functions are performed by hardware or software depends on particular applications and design constraint conditions of the technical solutions. A person skilled in the art may use different methods to implement the described functions for each particular application, but it should not be considered that the implementation goes beyond the scope of the present invention.

Steps of methods or algorithms described in the embodiments disclosed in this specification may be implemented by hardware, a software module executed by a processor, or a combination thereof. The software module may reside in a random access memory (RAM), a memory, a read-only memory (ROM), an electrically programmable ROM, an electrically erasable programmable ROM, a register, a hard disk, a removable disk, a CD-ROM, or any other form of storage medium known in the art.

In the foregoing specific implementations, the objective, technical solutions, and benefits of the present invention are further described in detail. It should be understood that the foregoing descriptions are merely specific implementations

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of the present invention, but are not intended to limit the protection scope of the present invention. Any modification, equivalent replacement, or improvement made without departing from the spirit and principle of the present invention should fall within the protection scope of the present invention.

What is claimed is:

1. An electronic device, comprising:
one or more processors; and
one or more memories coupled to the one or more processors and configured to store instructions that, when executed by the one or more processors, cause the electronic device to be configured to:
display a first interface of a first application, wherein the first interface comprises a first text content, and the first text content comprises a first text information;
copy the first text content into a clipboard in response to a first operation on the first interface, wherein the first text information is extracted from the first text content;
display, in response the first text information being a location and in response to a first condition being satisfied, a second interface of a second application when detecting an operation of a user opening the second application, wherein the second application is an electronic map, the first application differs from the second application, the second interface comprises a first option and an input box for receiving a pasted text to perform searching and location by the user, a position of the first option differs from a position of the input box, and the input box comprises a second text content indicating searching a location or a route by the user; and
perform a first function corresponding to the location in response to a second operation on the first option, wherein the first condition comprises a duration between a) the first operation and b) the operation of the user opening the second application being less than a preset amount of time.
2. The electronic device of claim 1, wherein when the first condition is not satisfied, the second interface is not displayed.
3. The electronic device of claim 1, wherein the first option comprises the first text information.
4. The electronic device of claim 1, wherein the first text content further comprises a second text information, and the second text information differs from the first text information.
5. The electronic device of claim 4, wherein the second text information comprises a metro line, and the second text information excludes the location.
6. The electronic device of claim 4, wherein the first text information and the second text information are continuous, with no overlap between the first text information and the second text information.
7. The electronic device of claim 1, wherein the instructions, when executed by the one or more processors, further cause the electronic device to be configured to:
detect a long pressing on the input box by the user;
display a paste option in response to the long pressing; and
paste the first text content in response to detecting a tapping on the paste option by the user.
8. The electronic device of claim 1, wherein the second operation is a tap operation on the first option.

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9. The electronic device of claim 1, wherein the first operation is a copy operation.

10. The electronic device of claim 1, wherein the instructions, when executed by the one or more processors, further cause the electronic device to be configured to:

display a second option on an interface of the second application, wherein the interface displaying the second option comprises the first text information; and
perform a second function corresponding to the location in response to a third operation on the second option, wherein the first function is locating or navigating to the location, the second function is locating or navigating to the location, and the second function differs from the first function.

11. The electronic device of claim 1, wherein the first text information is a keyword extracted from the first text content.

12. The electronic device of claim 1, wherein before displaying the second interface, the first option is displayed in response to the first text information being a location category, the second interface belongs to a first category, the first text information belongs to a second category, the first category is a location category, the second category is a location category, the first option is displayed when a matching result is not empty, and the matching result is from matching the first category against the second category.

13. The electronic device of claim 1, wherein the instructions, when executed by the one or more processors, further cause the electronic device to be configured to:

display a third option on an interface of the second application; and
ignore the first option in response to a fourth operation on the third option.

14. An electronic device, comprising:

one or more processors; and
one or more memories coupled to the one or more processors and configured to store instructions that, when executed by the one or more processors, cause the electronic device to be configured to:

display a first interface of a first application, wherein the first interface comprises a first text content, the first text content comprises a first text information and a second text information, the first text information is a location, and the first text information differs from the second text information;

copy the first text content into a clipboard in response to a first operation on the first interface;

display, in response to a duration between a) the first operation and b) an operation of a user opening a second application being less than a preset amount of time, a second interface of a second application when detecting the operation of the user opening the second application, wherein the second application is an electronic map, the first application differs from the second application, the second interface comprises a first option and an input box for receiving a pasted text to perform searching and location by the user, a position of the first option differs from a position of the input box, and the input box comprises a second text content indicating searching a location or a route by the user; and
perform a first function corresponding to the location in response to a second operation on the first option.

15. The electronic device of claim 14, wherein the first option comprises at least a part of the first text information automatically obtained.

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16. The electronic device of claim 15, wherein the at least a part of the first text information is an entirety of the first text information or a portion of the first text information.

17. The electronic device of claim 15, the at least a part of the first text information is a keyword extracted from the first text content.

18. The electronic device of claim 15, wherein before displaying the second interface, the first option is displayed in response to the at least a part of the first text information being a location category, the second interface belongs a first category, the first text information belongs a second category, the first category is a location category, the second category is a location category, and the first option is displayed in response to matching the first category against the second category.

19. The electronic device of claim 14, wherein the second text information comprises a metro line, and the second text information excludes the location.

20. The electronic device of claim 14, wherein the first text information and the second text information are continuous, with no overlap between the first text information and the second text information.

21. The electronic device of claim 14, wherein the instructions, when executed by the one or more processors, further cause the electronic device to be configured to:

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detect a long pressing on the input box by the user; display a paste option to be tapped by the user in response to the long pressing; and paste the first text content in response to detecting a tapping on the paste option by the user.

22. The electronic device of claim 14, wherein the second operation is a tap operation on the first option, and the first operation is a copy operation.

23. The electronic device of claim 14, wherein the instructions, when executed by the one or more processors, further cause the electronic device to be configured to:

display a second option on an interface of the second application, wherein the interface displaying the second option comprises the first text information; and perform a second function corresponding to the location in response to a third operation on the second option, wherein the first function is locating or navigating to the location, the second function is locating or navigating to the location, and the second function differs from the first function.

24. The electronic device of claim 14, wherein the instructions, when executed by the one or more processors, further cause the electronic device to be configured to:

display a third option on an interface of the second application; and ignore the first option in response to a fourth operation on the third option.

* * * * *